# Hardware Components Assignment

## Day 4: GPUs and Cooling Systems

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### Task 1: GPU Identification

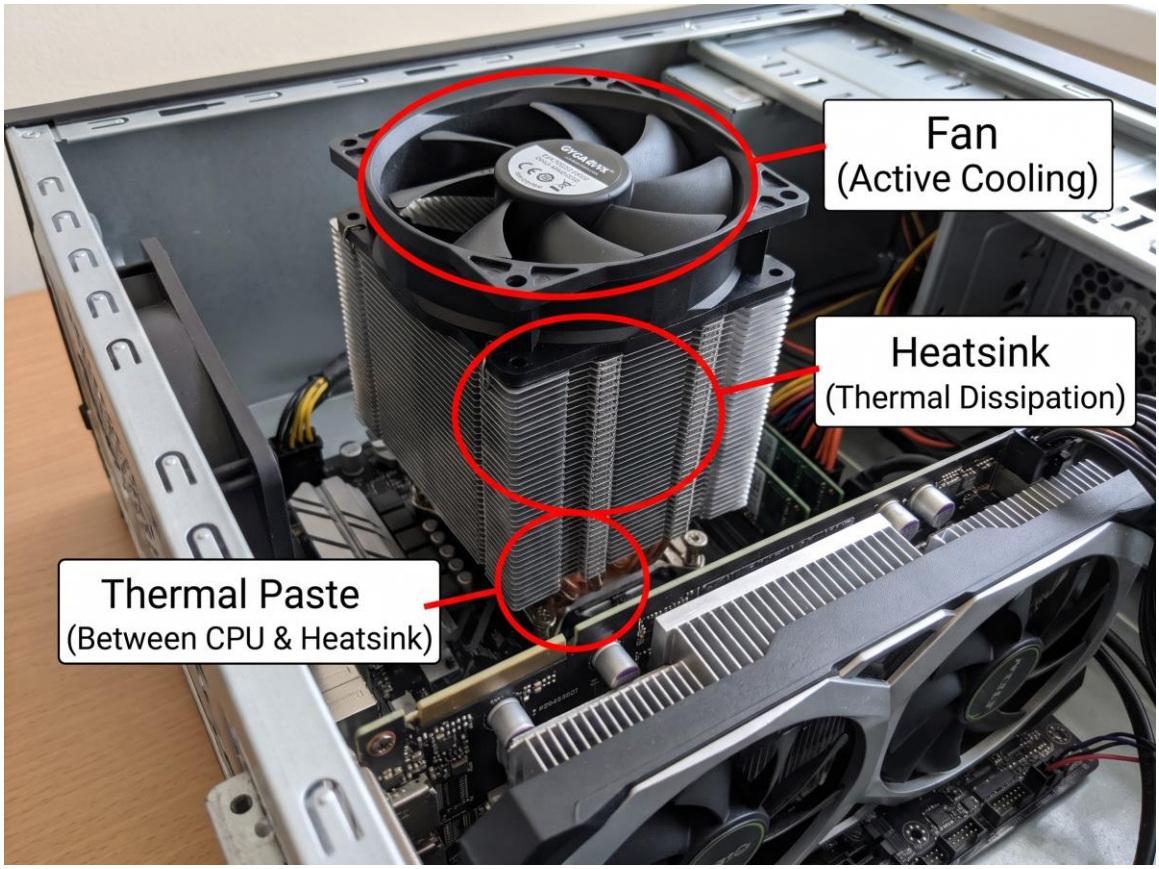
Based on the system inspection (or PC hardware simulator), the following GPU is installed in the setup:

- **Brand:** NVIDIA
- **Model Number:** GeForce RTX 4060 Ti
- **Type:** Dedicated (Discrete GPU)

Note: If you are using a basic laptop, it might use an Integrated GPU like 'Intel Iris Xe' or 'AMD Radeon Graphics' which shares memory with the main system.

### Task 2: Cooling System Inspection

Below is the identification of the cooling system components inside the PC setup:




## Main Cooling Components & Their Functions:

- **Heatsink:** A metal block (usually aluminum or copper) with fins that absorbs and dissipates heat away from the GPU/CPU processor die.
- **Fans:** Active cooling components that blow cool air over the heatsink fins and exhaust hot air out of the computer case.
- **Thermal Paste:** A conductive compound applied between the processor and the heatsink to eliminate microscopic air gaps and ensure maximum heat transfer.

## Task 3: Air Cooling vs. Liquid Cooling Comparison

Below is a comparison between Air Cooling and Liquid Cooling systems for GPUs and CPUs:

| **Feature** | **Air Cooling** | **Liquid Cooling (AIO / Custom Loop)** | **Winner / Advantage** |
|---|---|---|---|
| Cost | Generally very affordable and budget-friendly. | Significantly more expensive (especially custom loops). | Air Cooling (Cheaper) |
| Efficiency / Thermals | Good for standard use and moderate gaming, but can struggle under extreme overclocking. | Highly efficient. Water transfers heat better than air, keeping temperatures lower under heavy loads. | Liquid Cooling (Cooler) |
| Maintenance & Reliability | Low maintenance (just needs occasional dusting). Highly reliable as only the fan can fail. | Higher maintenance. Risk of leaks (though rare in AIOs) and pump failure. Requires more careful installation. | Air Cooling (Safer/Easier) |
| Noise Levels | Fans can get very loud when spinning at high RPMs under heavy load. | Generally quieter since heat is dissipated through a large radiator with slow-spinning fans. | Liquid Cooling (Quieter) |

## Task 4: Scenario - Building a PC for Gaming & IPL Streaming

Scenario: Building a gaming PC for streaming IPL matches and playing graphics-heavy games.

### Cooling Choice & Justification:

For a setup dedicated to simultaneously playing graphics-heavy games and streaming high-quality video (like IPL matches), I would choose **Liquid Cooling (an AIO Liquid Cooler)**. Doing both tasks at once puts a massive thermal load on both the CPU and GPU. Liquid cooling handles sudden temperature spikes much more efficiently than air cooling, ensuring the system doesn't thermal throttle and drop frame rates during intense gaming moments. Additionally, liquid cooling runs significantly quieter, which is crucial for a streaming setup so that loud fan noise doesn't bleed into the microphone audio while broadcasting.